



# TEST DOCUMENTATION REPORT

Wipro Capstone Project

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|---------------------|-------------------------------------------------|
| Project Title       | Wipro Capstone — <b>MakeMyTrip Train Module</b> |
| Project ID          | HarshVardhan_Singh_5381529_wipro_capstone       |
| Prepared By         | Harsh Vardhan Singh                             |
| Superset ID         | 5381529                                         |
| Program             | TRAINING (Selenium with Python B2)              |
| Date of Preparation | 23 May 2026                                     |
| Document Status     | Final                                           |

## 1. PROJECT DESCRIPTION

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### 1.1 Overview

This project automates the core **MakeMyTrip Train Booking** workflow, which integrates with the IRCTC system for train search, availability, passenger detail submission, and payment navigation. Built using Selenium WebDriver, Python, and Pytest, the framework strictly validates search listings, filter functionality, and UI input validations to ensure application reliability and execution efficiency.

Furthermore, the framework incorporates a Behavior-Driven Development (BDD) approach using Behave. By mapping technical test scripts to plain-English Gherkin scenarios, it creates accessible, living documentation that aligns automated test execution directly with business requirements

### 1.2 Test Strategy

#### 1.2.1 Test Strategy for Selenium Pytest

The testing strategy for this project is designed to validate the major functionalities of the MakeMyTrip Train Booking application through a structured and maintainable automation framework.

The framework uses Selenium WebDriver for browser automation and Pytest as the test execution framework. The Page Object Model (POM) design pattern is implemented to separate web element locators and page actions from test scripts, improving code maintainability, readability, and reusability.

Data-Driven Testing (DDT) is implemented using CSV and Excel files. CSV files are used for train search test data, while Excel files are used for passenger information. This approach allows execution of multiple test scenarios without modifying the automation scripts.

The framework also includes logging and screenshot capture utilities for better debugging, execution tracking, and reporting.

#### 1.2.1 Test Strategy for BDD Behave

The testing strategy for this project is designed to validate the major functionalities of the MakeMyTrip Train Booking application through a structured and highly readable behavior-driven automation framework.

The framework uses the Behave library as the core BDD execution engine to bridge the gap between business requirements and technical scripts. The Gherkin syntax (Given/When/Then) is implemented to separate plain-English test scenarios in

feature files from the underlying Python step definitions, improving team collaboration, readability, and requirement traceability.

Data-Driven Testing (DDT) is implemented using Gherkin Scenario Outlines alongside CSV and Excel files. CSV files are parsed for train search test data, while Excel files are consumed for passenger information directly within the step definitions. This approach allows execution of multiple test scenarios without modifying the core feature files or automation scripts.

The framework also includes custom environmental hooks (environment.py) alongside Allure integration and screenshot capture utilities for better debugging, execution tracking, and visually rich reporting.

## 2. TEST ENVIRONMENT

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### 2A. Selenium ,Pytest — Environment

- **Operating System:** Windows 11
- **IDE:** [ PyCharm]
- **Programming Language:** Python 3.11.1
- **Execution Framework:** Pytest
- **Automation Tool:** Selenium WebDriver (utilizing SeleniumBase for UC Mode and advanced browser handling)
- **Design Pattern:** Page Object Model (POM) for centralized web element management
- **Test Data Management:** External data injection using CSV / Excel readers (pandas or openpyxl) via Pytest parameterization
- **Dependencies:**

requirement.txt--

- Selenium
- webdriver-manager
- pytest
- Allure-pytest
- openpyxl
- pytest-html
- **Reporting & Logging:**
  - **Reports:** Allure HTML Reporting Framework (or Pytest-HTML)
  - **Logs:** Custom Python logging utility capturing detailed runtime traces and saving them to Logs/automation.log

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### 2B. BDD behave — Environment

- **Operating System:** Windows 11
- **IDE:** PyCharm

- **Programming Language:** Python 3.11.1
- **Execution Framework:** Behave (Behavior-Driven Development)
- **Automation Tool:** Selenium WebDriver (utilizing SeleniumBase for UC Mode)
- **Design Pattern:** Page Object Model (POM) injected dynamically via environment.py hooks
- **Test Data Management:** Gherkin Scenario Outlines combined with dynamic CSV / Excel reading within step definitions
- **Dependencies:**

#### requirements.txt--

- selenium
- behave
- webdriver-manager
- allure-behave
- pytest
- openpyxl
- configparser
- gherkin-official
- allure
- allure-python-commons
- **Reporting & Logging:**
  - **Reports:** Allure HTML Reporting generated via allure\_behave.formatter
  - **Logs:** Custom Python logging capturing runtime data, strictly attached to the Allure lifecycle within the after\_step and after\_scenario hooks

### 3. TEST EXECUTION

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#### 3A. Selenium + Pytest — Test Execution

##### *Test Summary*

| Metric              | Value                        |
|---------------------|------------------------------|
| Total Test Cases    | 7                            |
| Positive Test Cases | 4                            |
| Negative Test Cases | 2                            |
| End-to-End Flows    | 1                            |
| Test Execution Mode | Sequential — Chrome (headed) |
| Avg. Execution Time | ~4 minutes (full suite)      |

##### *End-to-End Test Flow*

The core End-to-End (E2E) test simulates a complete, real-world user journey for booking a train ticket. Driven by Pytest and parameterized using an external Excel dataset (test\_data.xlsx), the script executes a seamless 7-phase automation pipeline, capturing Allure screenshots and logging runtime data at every milestone.

##### **The 7-Phase Execution Pipeline:**

###### **Phase 1: Home Page Search**

- The framework reads the from\_station and to\_station from the Excel dataset.
- It inputs these values into the MakeMyTrip homepage and triggers the search algorithm.

###### **Phase 2: Train Listing & Dynamic Filters**

- Upon reaching the results page, the script verifies the DOM load.
- It applies critical business filters, specifically isolating AC Class and Available trains to ensure a valid booking path.

###### **Phase 3: Seat Selection**

- The framework intelligently locates and selects the first available train ticket matching the filter criteria.

###### **Phase 4: Passenger Data & Modals**

- The script handles complex UI elements, including selecting cancellation insurance and opening the dynamic "+ Add Traveller" modal.

- It injects specific passenger demographics (Name, Age, Gender) and contact details (Email, Mobile) directly from the Excel payload.

#### **Phase 5: IRCTC Account Verification**

- The system successfully interacts with the IRCTC authentication modal, injecting the required IRCTC\_ID.

#### **Phase 6: Final Booking Acknowledgement**

- The script checks mandatory terms and conditions checkboxes and clicks the final "Pay and Book Now" button to transition out of the booking phase.

#### **Phase 7: Secure Payment Gateway Integration**

- The framework navigates the payment sidebar to select the "Credit Card" option.
- It dynamically normalizes the Excel data keys to prevent payload errors and securely inputs the Card Number, Expiry, CVV, and Cardholder Name.
- **Final Assertion:** The test concludes with a strict Pytest assertion, scanning the URL path and DOM text to guarantee the driver successfully reached the secure checkout environment without stalling.

### ***Positive Test Cases (4)***

**1. Verify train search functionality using valid source and destination stations.** This test ensures that the core search engine accepts correct station data and initiates the query.

- **Validation:** It validates that the application successfully accepts the valid inputs via the HomePage POM methods without throwing exceptions and transitions the user away from the homepage.

**2. Validate that the train listing page displays available trains after successful search.** This scenario confirms that the application successfully retrieves and renders the timetable data from the database.

- **Validation:** It uses Pytest assertions to validate the presence and visibility of actual train listing web elements in the DOM, ensuring the results page is not blank.

**3. Verify train filters functionality using availability and class filters.** This test interacts with the sidebar filter checkboxes to narrow down a large list of train results.

- **Validation:** It validates that the UI dynamically updates to display only the trains that strictly match the selected criteria (e.g., filtering out Waitlisted or Non-AC trains) using TrainPage POM methods.

#### 4. Verify successful navigation to the payment page after passenger detail submission.

This simulates a user successfully completing the complex traveler details form and initiating the checkout process.

- **Validation:** It validates the state transition by asserting that the application's URL or DOM successfully shifts to the secure payment gateway environment after the Pay and Book Now button is clicked.

#### *Negative Test Cases (2)*

##### 1. Validate application behavior when invalid source or destination station is entered.

This test attempts to break the search by inputting non-existent, numeric, or gibberish station names.

- **Validation:** It validates through Pytest assertions that the system gracefully handles the bad data by blocking the search action and displaying an appropriate UI error, rather than crashing the application.

**2. Verify validation when invalid phone number is entered in passenger details.** This scenario attempts to bypass the contact form requirements by submitting an incomplete or malformed mobile number.

- **Validation:** It validates that the form submission is strictly blocked and asserts that a specific, red validation error text is rendered beneath the input field prompting the user to fix the data.

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### 3B. BDD behave — Test Execution

#### *Test Summary*

| Metric              | Value                     |
|---------------------|---------------------------|
| Total Scenarios     | 7                         |
| Positive Scenarios  | 4                         |
| Negative Scenarios  | 2                         |
| End-to-End Flows    | 1                         |
| Feature Files       | 2                         |
| Avg. Execution Time | ~4.5 minutes (full suite) |

#### *End-to-End Scenario*



**Overview** The core End-to-End (E2E) scenario validates the complete train ticket booking journey through the lens of Behavior-Driven Development (BDD). Written in plain-English Gherkin syntax, the scenario acts as living documentation while executing a complex, data-driven automation pipeline powered by Behave, Selenium, and external Excel test data.

### **The Execution Pipeline:**

#### **Phase 1: Application Launch & Data Ingestion**

- **Gherkin Steps:** Given User launches MakeMyTrip application, and User loads test data from Excel "test\_data.xlsx".
- **Action:** The framework initializes the SeleniumBase UC driver, navigates to the base URL, and reads the external Excel payload into the Behave context object for dynamic use throughout the scenario.

#### **Phase 2: Navigation & Train Search**

- **Gherkin Steps:** When User closes the login popup, and User navigates to the Trains section. Then the train search dashboard should be visible. When User enters source station from Excel, and User enters destination station from Excel, and User selects a travel date, and User selects the class type, and User clicks on Search Trains button.
- **Action:** The script handles initial UI blockers (popups), navigates to the Train Booking module, and inputs the source station, destination station, travel date, and class type using the previously loaded Excel data.

#### **Phase 3: Results Validation & Dynamic Filtering**

- **Gherkin Steps:** Then the train listing results page should be loaded. When User applies the AC and Available filters, and User selects the first available train ticket.
- **Action:** A strict assertion verifies the successful rendering of the search results DOM. The script then applies business-critical filters (AC Class & Available Status) and selects the first qualifying train ticket.

#### **Phase 4: Passenger Demographics & IRCTC Verification**

- **Gherkin Steps:** Then the specific train details page should be loaded. When User opens the add traveller modal and enters passenger details from Excel, and User enters IRCTC username from Excel, and User fills contact details from Excel.
- **Action:** The script navigates the complex specific train details page, interacting with pop-up modals to inject passenger details, IRCTC credentials, and contact information strictly pulled from the external dataset.

#### **Phase 5: Booking Acknowledgement & Navigation**

- **Gherkin Steps:** And User selects cancellation insurance and accepts mandatory terms, and User proceeds to payment.

- **Action:** The framework completes the mandatory UI checkboxes and triggers the final progression to route the user out of the booking environment and into the checkout environment.

## **Phase 6: Secure Payment Gateway Integration**

- **Gherkin Steps:** Then the payment page should be loaded and credit card option selected. When User enters credit card details from Excel, then the payment continue button should become active.
- **Action:** The final phase validates that the application successfully reaches the payment gateway. It selects the credit card option, securely inputs the card details from the Excel payload, and asserts that the final payment confirmation button becomes active and ready for transaction processing.

## *Positive Scenarios (4)'*

**1. Verify train search functionality using valid source and destination stations.** This test ensures that the core search engine correctly processes a standard user query.

- **Validation:** It validates that when a user inputs a valid source station, destination station, and travel date from the test data (CSV), the system successfully executes the search and loads the results dashboard.

**2. Validate that the train listing page displays available trains after successful search.** This scenario confirms the retrieval and rendering of database information.

- **Validation:** It validates that after clicking the search button, the application successfully populates the results page with actual, visible train listings rather than returning a blank page or an error.

**3. Verify train filters functionality using availability and class filters.** This test checks the dynamic sorting and filtering logic of the UI.

- **Validation:** It validates that when a user applies specific constraints—such as "AC" class and "Available" status—the train listing correctly updates to display only the results that strictly match those filters.

**4. Verify successful navigation to the payment page after passenger detail submission.** This scenario tests the complex data-entry and state transition of the booking flow.

- **Validation:** It validates that a user can successfully select a train, populate the add-traveler modal, input an IRCTC ID, provide valid contact details, accept terms, and be securely routed to the payment gateway without any system interruptions.

### *Negative Scenarios (2)*

**1. Validate application behavior when the same source and destination stations are entered.** This test verifies the application's logical validation regarding geography.

- **Validation:** It validates that the system correctly identifies when a user attempts to search with identical source and destination stations, preventing the search and displaying a specific error message stating stations cannot be the same.

**2. Verify validation when invalid phone number is entered in passenger details.** This scenario checks the form-field security and data integrity validation.

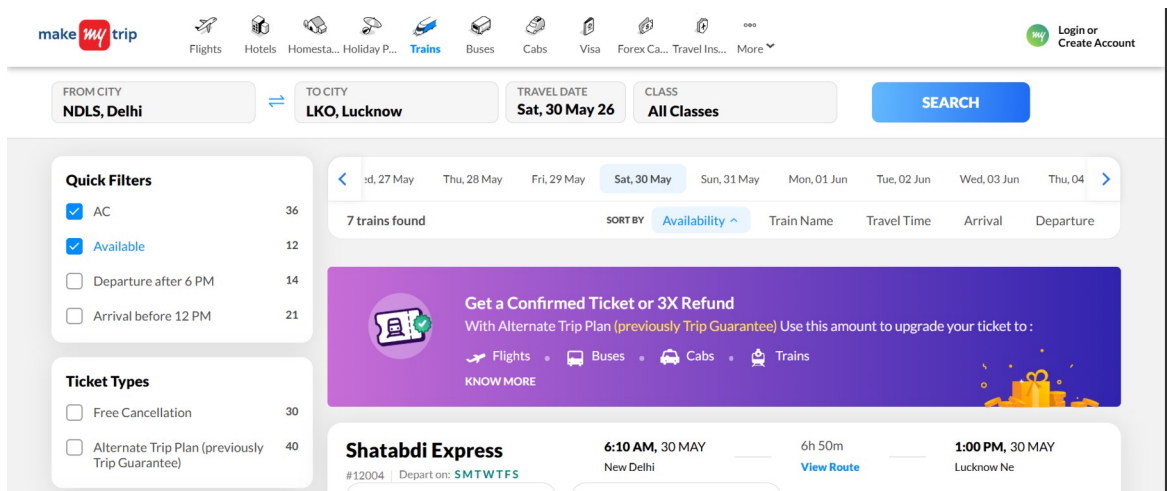
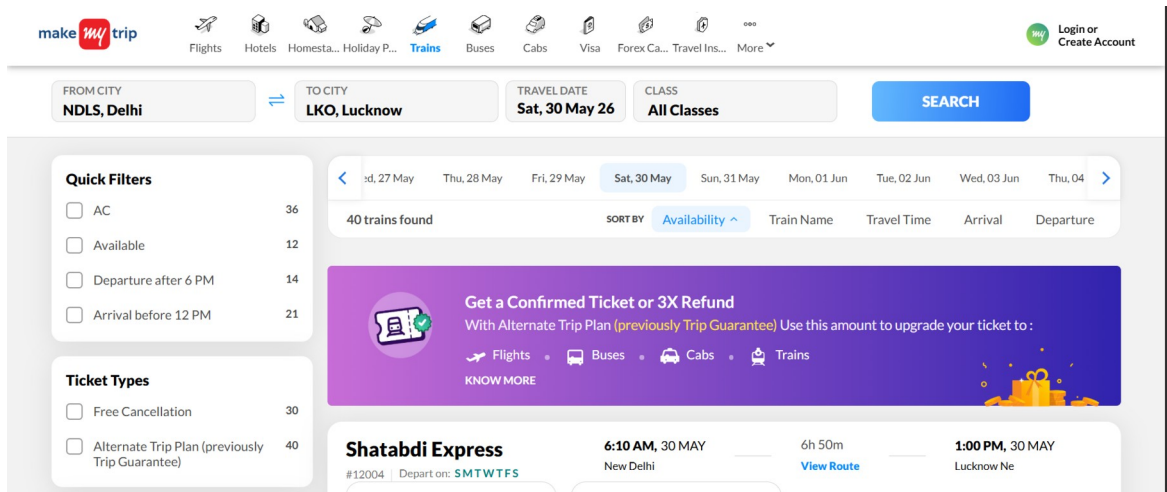
- **Validation:** It validates that if a user attempts to proceed to payment with a malformed or invalid mobile number, the application strictly blocks the submission and displays a clear validation error on the UI prompting the user to correct the contact details.

## 4. TEST OUTPUTS

### 4A. Selenium + pytest — Test Outputs

#### 4A.1 Screenshots

Screenshots are automatically captured by the pytest confest.py fixture on both pass and fail events. Files are saved to the screenshots/ directory with a timestamp-prefixed filename for traceability.



## Select Travellers

## Shatabdi Express

#12004 | Departs on: SMTWTF5

6:10 AM, 30 May — 6 hrs 50 mins — 1:00 PM, 30 May  
New Delhi (NDLS) Lucknow Ne (LJN)

## Availability Status

CC AVAILABLE-0636

Updated few mins ago

## Your Boarding Station

NEW DELHI (NDLS) - 6:10 AM (30 May)

Change

**Get full ticket fare refund on Cancellation**  
Just at ₹215 per person

Zero charges when the ticket is cancelled

Zero Convenience  
fees for logged-in  
users

Login

## PAY &amp; BOOK NOW

|                     |      |
|---------------------|------|
| Base fare per adult | ₹715 |
| Dynamic fare        | ₹286 |
| Catering charge     | ₹125 |
| Reservation charge  | ₹40  |
| Superfast charge    | ₹45  |

## IRCTC Password

Please ensure you have your password ready. In case you don't have your password, we strongly recommend that you get a new password now.

Get New Password

Zero Convenience  
fees for logged-in  
users

Login

## PAY &amp; BOOK NOW

|                     |      |
|---------------------|------|
| Base fare per adult | ₹715 |
| Dynamic fare        | ₹286 |
| Catering charge     | ₹125 |
| Reservation charge  | ₹40  |
| Superfast charge    | ₹45  |

**Total Price per adult ₹1270**

## Contact Information (We'll send your ticket and status updates here)

## Email Id

example@test.com

## Mobile Number

9876543210

## Offers &amp; Discounts



HAVE A COUPON CODE?

APPLY

## Offers &amp; Discounts



HAVE A COUPON CODE?

APPLY

## Additional Preferences Optional

Your  
State

(Required for GST purpose on your tax invoice. You can edit this anytime later in your profile section.)

Select the State

Uttar Pradesh

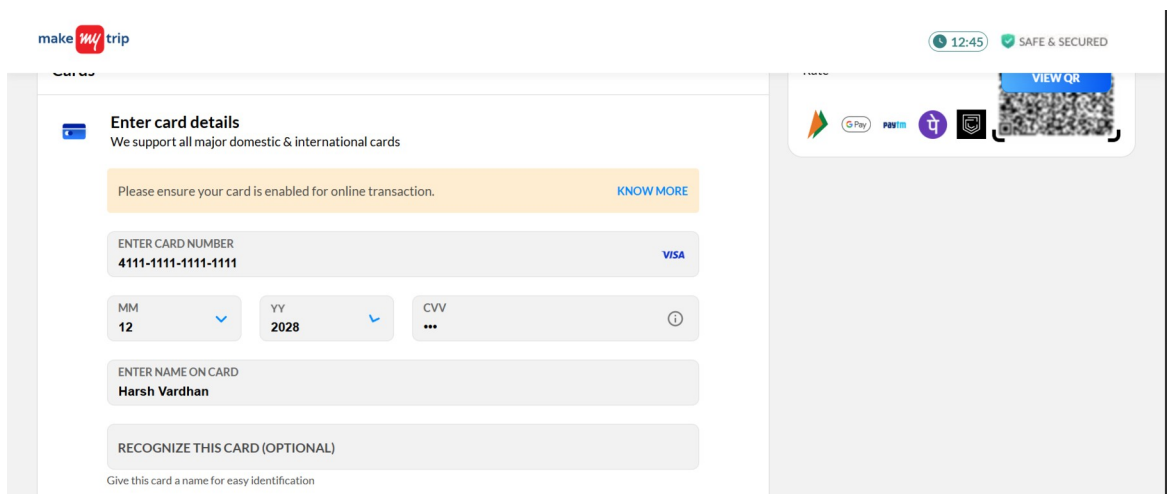
☒ Confirm and save billing details to your profileZero Convenience  
fees for logged-in  
users

Login

## PAY &amp; BOOK NOW

|                     |      |
|---------------------|------|
| Base fare per adult | ₹715 |
| Dynamic fare        | ₹286 |
| Catering charge     | ₹125 |
| Reservation charge  | ₹40  |
| Superfast charge    | ₹45  |

**Total Price per adult ₹1270**



#### 4A.2 Log File — Sample (logs/automation.log)

```

2026-05-19 22:30:53,183 - INFO - =====
2026-05-19 22:30:53,183 - INFO - Starting Test
2026-05-19 22:30:53,183 - INFO - Browser : chrome
2026-05-19 22:30:53,184 - INFO - Base URL :
https://www.makemytrip.com/railways/
2026-05-19 22:30:53,184 - INFO - Implicit Wait : 5
2026-05-19 22:30:53,184 - INFO - Timeout : 10
2026-05-19 22:30:53,184 - INFO - Headless : False
2026-05-19 22:30:53,204 - INFO - ===== WebDriver manager =====
2026-05-19 22:30:54,242 - INFO - Driver [C:
\Users\harsh\.wdm\drivers\chromedriver\win64\148.0.7778.167\chromedriver-
win64\chromedriver.exe] found in cache by browser version
2026-05-19 22:30:58,314 - INFO - Opening URL :
https://www.makemytrip.com/railways/
2026-05-19 22:30:58,739 - INFO - STARTING TRAIN SEARCH FLOW FOR: Delhi TO
Lucknow
2026-05-19 22:30:58,740 - INFO - STARTING TRAIN SEARCH
2026-05-19 22:31:04,728 - INFO - LOGIN POPUP CLOSED
2026-05-19 22:31:07,750 - INFO - FROM STATION: Delhi
2026-05-19 22:31:07,947 - INFO - CLICKED FIELD: Delhi
2026-05-19 22:31:11,196 - INFO - TYPED STATION: Delhi
2026-05-19 22:31:14,335 - INFO - SELECTED STATION: Delhi
2026-05-19 22:31:16,337 - INFO - TO STATION: Lucknow
2026-05-19 22:31:16,379 - INFO - CLICKED FIELD: Lucknow
2026-05-19 22:31:19,595 - INFO - TYPED STATION: Lucknow
2026-05-19 22:31:22,645 - INFO - SELECTED STATION: Lucknow
2026-05-19 22:31:24,685 - INFO - DATE BOX CLICKED
2026-05-19 22:31:26,720 - INFO - 30 MAY 2026 DATE SELECTED
2026-05-19 22:31:28,721 - INFO - CLICKING SEARCH BUTTON
2026-05-19 22:31:32,586 - INFO - SEARCH BUTTON CLICKED

```

2026-05-19 22:31:42,596 - INFO - TRAIN SEARCH COMPLETED  
2026-05-19 22:31:42,596 - INFO - TRAIN SEARCH COMPLETED FROM HOME PAGE  
2026-05-19 22:31:42,761 - INFO - Screenshot saved at:  
reports/screenshots/1\_After\_Homepage\_Search\_20260519\_223142.png  
2026-05-19 22:31:42,777 - INFO - POM LOG: VERIFYING TRAIN SEARCH RESULTS  
PAGE  
2026-05-19 22:31:44,796 - INFO - POM LOG: TRAIN SEARCH RESULTS CONTAINER  
VALIDATED  
2026-05-19 22:31:44,796 - INFO - POM LOG: APPLYING AC FILTER  
2026-05-19 22:31:46,013 - INFO - POM LOG: AC FILTER DEPLOYED NATIVELY  
2026-05-19 22:31:49,018 - INFO - POM LOG: APPLYING AVAILABILITY FILTER  
2026-05-19 22:31:50,144 - INFO - POM LOG: AVAILABILITY FILTER DEPLOYED  
NATIVELY  
2026-05-19 22:31:53,597 - INFO - Screenshot saved at:  
reports/screenshots/2\_After\_Applying\_Filters\_20260519\_223153.png  
2026-05-19 22:31:53,621 - INFO - POM LOG: SELECTING FIRST AVAILABLE TRAIN  
SEAT OPTION  
2026-05-19 22:31:56,657 - INFO - POM LOG: VERIFYING TARGET ELEMENT TEXT  
---> 'Available 636'  
2026-05-19 22:31:58,303 - INFO - POM LOG: NATIVE CLICK SENT TO  
AVAILABILITY TEXT ELEMENT  
2026-05-19 22:32:03,310 - INFO - TRAIN SPECIFIC RUN SELECTION CAPTURED  
SUCCESSFULLY  
2026-05-19 22:32:21,479 - INFO - Screenshot saved at:  
reports/screenshots/3\_After\_Seat\_Selection\_20260519\_223203.png  
2026-05-19 22:32:21,542 - INFO - POM LOG: TARGETING CANCELLATION  
PROTECTION RADIO BUTTON  
2026-05-19 22:32:23,136 - INFO - POM LOG: CANCELLATION SELECTION  
SUCCESSFULLY COMPLETED  
2026-05-19 22:32:23,137 - INFO - POM LOG: DISPATCHING CLICK INTERACTION  
TO ADD TRAVELLER LINK  
2026-05-19 22:32:24,730 - INFO - POM LOG: ADD TRAVELLER POPUP MODAL  
INSTANTIATED SUCCESSFULLY  
2026-05-19 22:32:24,731 - INFO - POM LOG: FILLING MODAL -> NAME: Harsh,  
AGE: 25, GENDER: Male  
2026-05-19 22:32:32,005 - INFO - POM LOG: TRAVELLER SUBMITTED  
SUCCESSFULLY  
2026-05-19 22:32:32,006 - INFO - TRAVELLER POPULATED SUCCESSFULLY VIA  
EXCEL  
2026-05-19 22:32:32,006 - INFO - POM LOG: FILLING CONTACT DETAILS ->  
EMAIL: example@test.com, MOBILE: 9876543210  
2026-05-19 22:32:32,996 - INFO - POM LOG: CONTACT DETAILS POPULATED  
SUCCESSFULLY  
2026-05-19 22:32:32,996 - INFO - CONTACT DETAILS POPULATED SUCCESSFULLY  
2026-05-19 22:32:33,095 - INFO - Screenshot saved at:  
reports/screenshots/4\_After\_Passenger\_Details\_20260519\_223232.png  
2026-05-19 22:32:33,123 - INFO - POM LOG: INITIATING IRCTC USERNAME  
OVERLAY SELECTION FOR: abhinavy918  
2026-05-19 22:32:38,739 - INFO - POM LOG: Modal not detected. Forcing

click on IRCTC trigger via JavaScript.

2026-05-19 22:32:45,858 - INFO - POM LOG: Clicking explicitly inside the IRCTC username popup input field.

2026-05-19 22:32:46,861 - INFO - POM LOG: IRCTC ID TYPED SUCCESSFULLY INTO ACTIVE MODAL INPUT LAYER

2026-05-19 22:32:50,913 - INFO - POM LOG: IRCTC ID SUBMITTED VIA MODAL EXECUTION

2026-05-19 22:32:50,913 - INFO - END-TO-END FLOW RUN COMPLETED SUCCESSFULLY

2026-05-19 22:32:51,009 - INFO - Screenshot saved at: reports/screenshots/5\_After\_IRCTC\_Entry\_20260519\_223250.png

2026-05-19 22:32:51,025 - INFO - POM LOG: Targeting the mandatory confirmation checkbox wrapper.

2026-05-19 22:32:51,562 - INFO - POM LOG: Mandatory checkbox successfully marked.

2026-05-19 22:32:52,573 - INFO - MANDATORY ACKNOWLEDGEMENT CHECKBOX CONFIRMED AND SUCCESSFUL

2026-05-19 22:32:52,595 - INFO - POM LOG: Targeting the 'PAY & BOOK NOW' button anchor link.

2026-05-19 22:32:53,152 - INFO - POM LOG: 'PAY & BOOK NOW' action dispatched successfully.

2026-05-19 22:32:53,156 - INFO - PAY AND BOOK NOW BUTTON SELECTION COMPLETED SUCCESSFULLY

2026-05-19 22:32:53,235 - INFO - Screenshot saved at: reports/screenshots/6\_After\_Clicking\_Pay\_And\_Book\_20260519\_223253.png

2026-05-19 22:32:53,254 - INFO - POM LOG: LOCATING CREDIT/DEBIT CARD PAYMENT TAB

2026-05-19 22:33:00,845 - INFO - POM LOG: CREDIT/DEBIT CARD TAB SELECTED SUCCESSFULLY

2026-05-19 22:33:02,849 - INFO - POM LOG: INITIALIZING CARD DETAILS INJECTION

2026-05-19 22:33:03,056 - INFO - POM LOG: CARD NUMBER COMMITTED

2026-05-19 22:33:04,126 - INFO - POM LOG: EXPIRY MONTH COMMITTED -> 12

2026-05-19 22:33:04,701 - INFO - POM LOG: EXPIRY YEAR COMMITTED -> 28

2026-05-19 22:33:05,291 - INFO - POM LOG: CVV COMMITTED

2026-05-19 22:33:05,903 - INFO - POM LOG: CARD HOLDER NAME COMMITTED

2026-05-19 22:33:06,904 - INFO - POM LOG: ALL DUMMY PAYMENT DETAILS INJECTED SUCCESSFULLY. END OF E2E FLOW REACHED.

2026-05-19 22:33:09,905 - INFO - TEST PIPELINE: PAYMENT DATA COMPLETELY COMMITTED

2026-05-19 22:33:10,022 - INFO - Screenshot saved at: reports/screenshots/7\_Final\_Payment\_Page\_State\_20260519\_223309.png

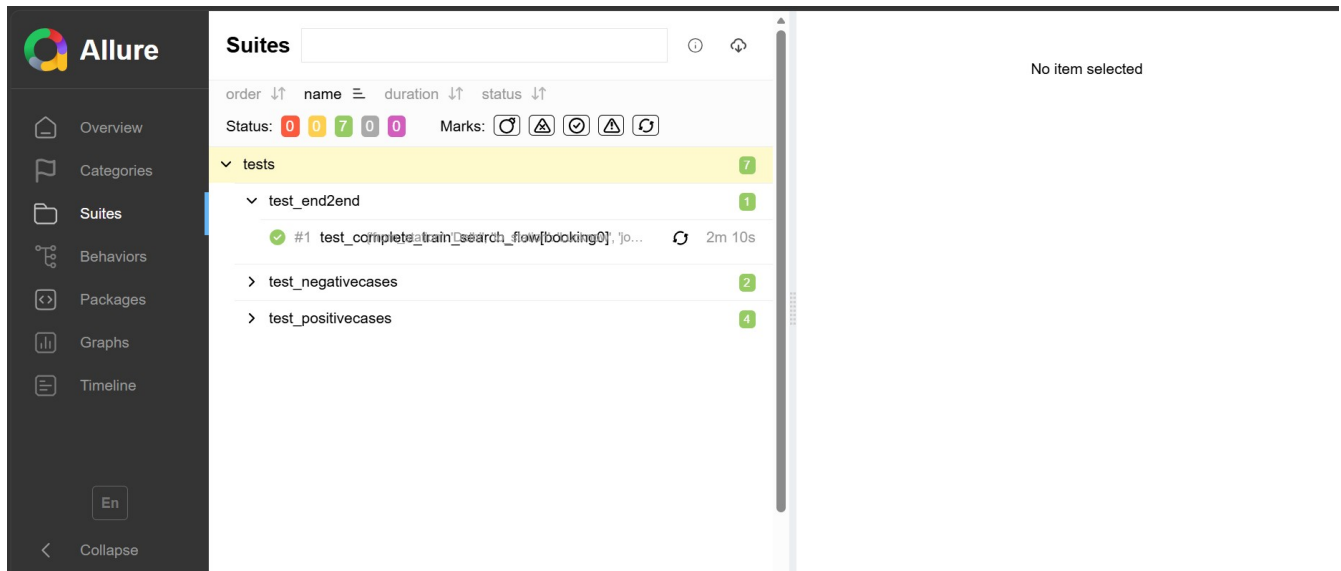
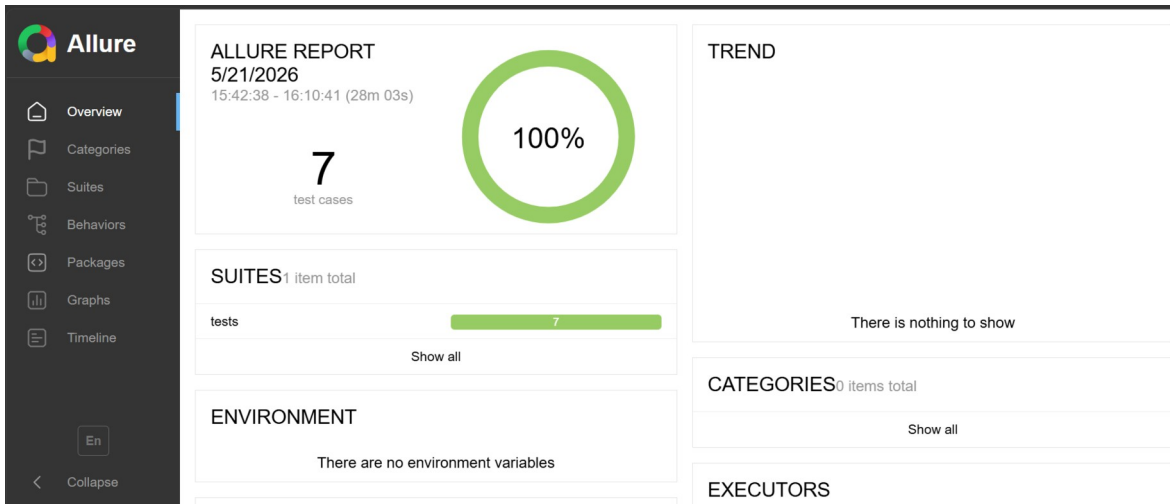
2026-05-19 22:33:10,126 - INFO - Closing Browser

2026-05-19 22:33:21,519 - INFO - Ending Test

2026-05-19 22:33:21,519 - INFO - =====



### 4A.3 Allure Report



Allure

Overview

Categories

Suites

Behaviors

Packages

Graphs

Timeline

En

Collapse

test\_end2end1

#1 test\_complete(train\_search\_flow(booking0)Lucknow', 'journey\_\_': datetime.datetime(2026, 5, 30, 0, 0), 'Passenger\_Name': '...

2m 10s

test\_negativcases2

#1 NS\_01: Validate Invalid Station Search Error Handling('test\_id': 'NEG\_01', 'scenario\_type': 'invalid\_station', 'from\_station': 'XYZ\_FAKE\_STATION', 'to\_station': 'NDL...

39s 335ms

#2 NS\_02: Verify Invalid Mobile Field Form Validation('test\_id': 'NEG\_02', 'scenario\_type': 'invalid\_phone', 'from\_station': 'NDLS', 'to\_station': 'AYC', 'Passenger\_Name': 'H...

1m 05s

test\_positivcases4

#1 TS\_01: Verify Train Search Functionality('from\_station': 'Delhi', 'to\_station': 'Ayodhya', 'Passenger\_Name': 'Harsh Vardhan', 'Passenger\_Age': '28', 'Passenger...

48s 874ms

#2 TS\_02: Validate Train Listing Display('from\_station': 'Delhi', 'to\_station': 'Ayodhya', 'Passenger\_Name': 'Harsh Vardhan', 'Passenger\_Age': '28', 'Passenger\_...

43s 584ms

#3 TS\_03: Verify Train Filters Functionality('from\_station': 'Delhi', 'to\_station': 'Ayodhya', 'Passenger\_Name': 'Harsh Vardhan', 'Passenger\_Age': '28', 'Passenger\_...

52s 413ms

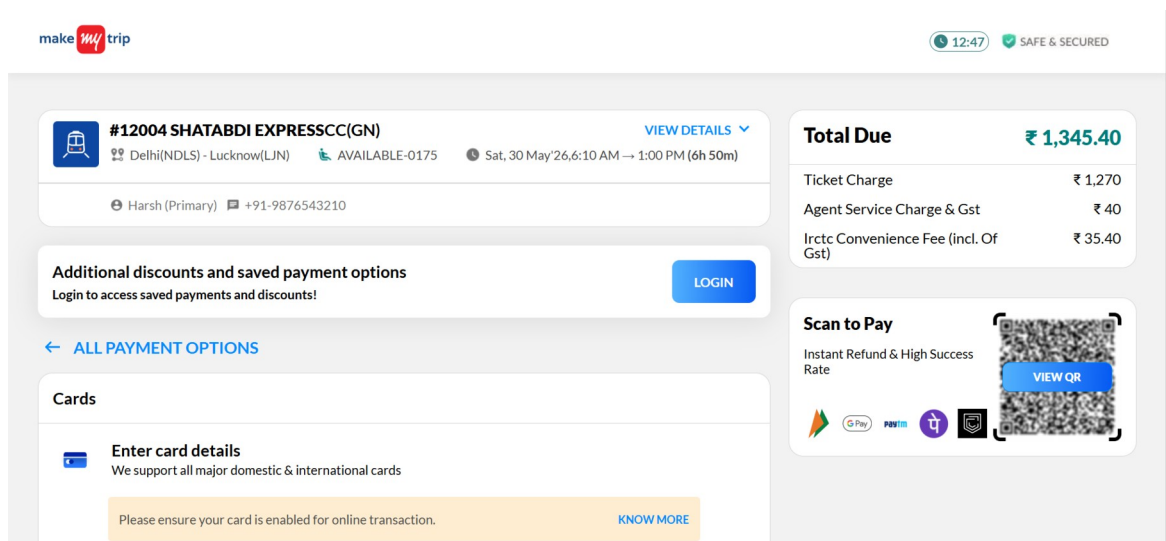
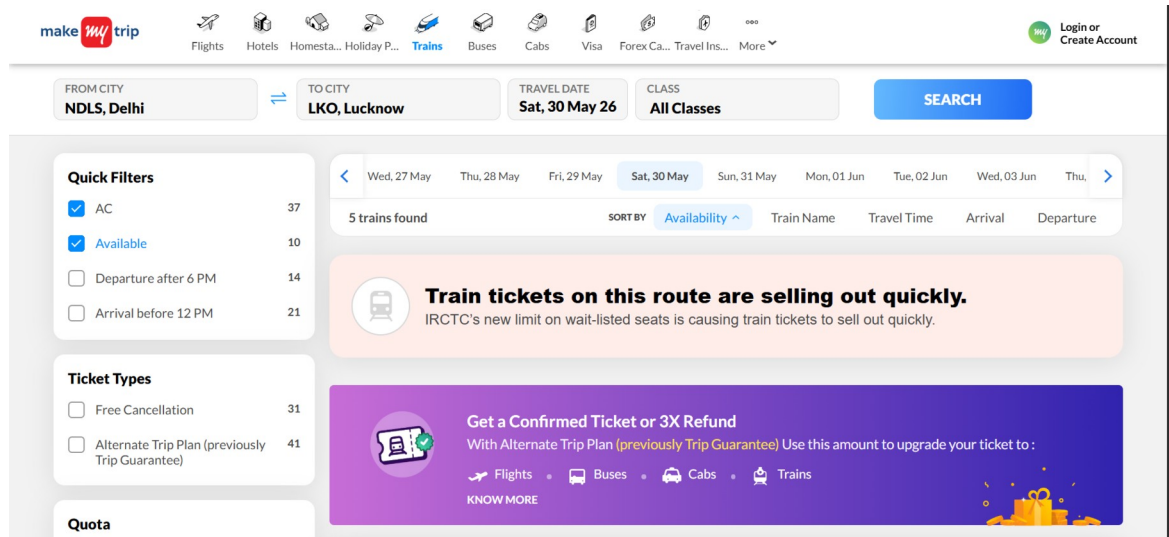
#4 TS\_04: Verify Successful('from\_station': 'Delhi', 'to\_station': 'Ayodhya', 'Passenger\_Name': 'Harsh Vardhan', 'Passenger\_Age': '28', 'Passenge...

1m 48s

No item selected

## 4B. BDD behave — Test Outputs

### 4B.1 Screenshots



make my trip

Flights

Hotels

Homesta... Holiday P...

Trains

Buses

Cabs

Visa

Forex Ca...

Travel Ins...

More

my

Login or Create Account

FROM CITY

NDLS, Delhi

TO CITY

LKO, Lucknow

TRAVEL DATE

Sat, 30 May 26

CLASS

All Classes

SEARCH

Quick Filters

☐ AC

37

☐ Available

10

☐ Departure after 6 PM

14

☐ Arrival before 12 PM

21

Ticket Types

☐ Free Cancellation

31

☐ Alternate Trip Plan (previously Trip Guarantee)

41

Quota

<

Wed, 27 May

Thu, 28 May

Fri, 29 May

Sat, 30 May

Sun, 31 May

Mon, 01 Jun

Tue, 02 Jun

Wed, 03 Jun

Thu,

>

41 trains found

SORT BY

Availability

Train Name

Travel Time

Arrival

Departure

**Train tickets on this route are selling out quickly.**  
IRCTC's new limit on wait-listed seats is causing train tickets to sell out quickly.

**Get a Confirmed Ticket or 3X Refund**  
With Alternate Trip Plan (previously Trip Guarantee) Use this amount to upgrade your ticket to :  
✈ Flights • 🚌 Buses • 🚗 Cabs • 🚂 Trains  
KNOW MORE

Flights

Hotels

Villas & Homestays

Holiday Packages

Trains

Buses

Cabs

Tours & Attractions

Visa

Cruise

Forex Card & Currency

Travel Insurance

Book Train Tickets

Check PNR Status

Live Train Status

Train Ticket Booking  
IRCTC Authorized e-ticketing

From

Delhi

NDLS, New Delhi Railway Station

To

Delhi

NDLS, New Delhi Railway Station

Travel Date

24 May '26

Sunday

Class

ALL

All Class

From & To station cannot be the same

SEARCH

Offers

Trains

All Offers

Hotels

Flights

Holidays

Bus

Cabs

VIEW ALL

<

>

**For Your Char Dham Journey: Up to 40% OFF\***  
on stays, packages, buses, cabs, trains & flights.

T&C'S APPLY

**Get a Confirmed Ticket or a 3x Refund**  
with our Alternate Trip Plan Feature

T&C'S APPLY

**INT For**  
and

#### 4B.2 Log File — Sample (logs/automation.log)

```
2026-05-23 11:22:51,731 - INFO - =====
2026-05-23 11:22:51,732 - INFO - AUTOMATION EXECUTION STARTED
2026-05-23 11:22:51,732 - INFO - Deleting old allure-results folder
2026-05-23 11:22:51,829 - INFO - Starting Behave Test Execution
2026-05-23 11:22:56,134 - INFO - ===== STARTING SCENARIO:
Successfully search, select, and enter payment details for a train ticket
=====
2026-05-23 11:23:10,834 - INFO - Implicit wait fetched : 10
2026-05-23 11:23:20,430 - INFO - POM LOG: BasePage initialized
2026-05-23 11:23:20,435 - INFO - POM LOG: BasePage initialized
2026-05-23 11:23:20,436 - INFO - POM LOG: BasePage initialized
2026-05-23 11:23:20,437 - INFO - POM LOG: BasePage initialized
2026-05-23 11:23:22,523 - INFO - CURRENT URL:
https://www.makemytrip.com/railways/
2026-05-23 11:23:22,718 - INFO - Application launched successfully
2026-05-23 11:23:22,719 - INFO - STEP PASSED: User launches MakeMyTrip
application - CAPTURING SCREENSHOT
2026-05-23 11:23:23,218 - INFO - Screenshot saved at:
reports/screenshots/PASS_User_launches_MakeMy_20260523_112322.png
2026-05-23 11:23:23,802 - INFO - Loading test data from:
testdata\test_data.xlsx
2026-05-23 11:23:24,224 - INFO - Data successfully loaded:
{'from_station': 'Delhi', 'to_station': 'Lucknow', 'journey__':
datetime.datetime(2026, 5, 30, 0, 0), 'Passenger_Name': 'Harsh',
'Passenger_Age': 25, 'Passenger_Gender': 'Male', 'Berth_Preference':
'N/A', 'Email': 'example@test.com', 'Mobile_Number': 9876543210,
'IRCTC_ID': 'abhinavy918', 'CARD_NUMBER': 4111111111111111, 'EXPIRY':
'12/28', 'CVV': 123, 'CARD HOLDER': 'Harsh Vardhan'}
2026-05-23 11:23:24,310 - INFO - Loaded test data for passenger: Harsh
2026-05-23 11:23:24,310 - INFO - STEP PASSED: User loads test data from
Excel "test_data.xlsx" - CAPTURING SCREENSHOT
2026-05-23 11:23:24,804 - INFO - Screenshot saved at:
reports/screenshots/PASS_User_loads_test_data_20260523_112324.png
2026-05-23 11:23:25,202 - INFO - POM LOG: Attempting to handle potential
login popup modal
2026-05-23 11:23:27,332 - INFO - POM LOG: LOGIN POPUP SUCCESSFULLY CLOSED
VIA XPATH CLICK
2026-05-23 11:23:28,339 - INFO - STEP PASSED: User closes the login popup
- CAPTURING SCREENSHOT
2026-05-23 11:23:28,640 - INFO - Screenshot saved at:
reports/screenshots/PASS_User_closes_the_logi_20260523_112328.png
2026-05-23 11:23:29,007 - INFO - POM LOG: Initializing selection of the
Trains navigation tab
2026-05-23 11:23:29,013 - INFO - STEP PASSED: User navigates to the
Trains section - CAPTURING SCREENSHOT
2026-05-23 11:23:29,394 - INFO - Screenshot saved at:
```

```
reports/screenshots/PASS_User_navigates_to_th_20260523_112329.png
2026-05-23 11:23:29,642 - INFO - Train search dashboard is visible
2026-05-23 11:23:29,642 - INFO - STEP PASSED: the train search dashboard
should be visible - CAPTURING SCREENSHOT
2026-05-23 11:23:29,876 - INFO - Screenshot saved at:
reports/screenshots/PASS_the_train_search_das_20260523_112329.png
2026-05-23 11:23:30,096 - INFO - POM LOG: Attempting to select station:
Delhi
2026-05-23 11:23:33,435 - INFO - STEP PASSED: User enters source station
from Excel - CAPTURING SCREENSHOT
2026-05-23 11:23:33,820 - INFO - Screenshot saved at:
reports/screenshots/PASS_User_enters_source_s_20260523_112333.png
2026-05-23 11:23:34,184 - INFO - POM LOG: Attempting to select station:
Lucknow
2026-05-23 11:23:35,415 - INFO - POM LOG: Label click intercepted,
attempting direct input interaction.
2026-05-23 11:23:37,849 - INFO - STEP PASSED: User enters destination
station from Excel - CAPTURING SCREENSHOT
2026-05-23 11:23:38,178 - INFO - Screenshot saved at:
reports/screenshots/PASS_User_enters_destinat_20260523_112337.png
2026-05-23 11:23:38,463 - INFO - POM LOG: Starting target travel date
selection sequence
2026-05-23 11:23:38,624 - INFO - POM LOG: Travel date calendar container
expanded
2026-05-23 11:23:41,401 - INFO - POM LOG: 30 MAY 2026 DATE SELECTION
SUBMITTED SUCCESSFULLY
2026-05-23 11:23:43,402 - INFO - STEP PASSED: User selects a travel date
- CAPTURING SCREENSHOT
2026-05-23 11:23:43,618 - INFO - Screenshot saved at:
reports/screenshots/PASS_User_selects_a_trave_20260523_112343.png
2026-05-23 11:23:43,817 - INFO - POM LOG: Initiating Travel Class
dropdown selector configuration
2026-05-23 11:23:43,868 - INFO - POM LOG: Travel Class dropdown opened
2026-05-23 11:23:45,987 - INFO - POM LOG: ALL CLASS FILTER SELECTED
SUCCESSFULLY
2026-05-23 11:23:48,281 - INFO - STEP PASSED: User selects the class type
- CAPTURING SCREENSHOT
2026-05-23 11:23:49,171 - INFO - Screenshot saved at:
reports/screenshots/PASS_User_selects_the_cla_20260523_112348.png
2026-05-23 11:23:49,670 - INFO - POM LOG: Dispatching form submission
execution via Search Button wrapper link
2026-05-23 11:23:55,214 - INFO - POM LOG: SEARCH ACTION EVENT SENT TO
CORE ENGINE SUCCESSFULLY
2026-05-23 11:24:00,214 - INFO - STEP PASSED: User clicks on Search
Trains button - CAPTURING SCREENSHOT
2026-05-23 11:24:00,781 - INFO - Screenshot saved at:
reports/screenshots/PASS_User_clicks_on_Searc_20260523_112400.png
2026-05-23 11:24:01,814 - INFO - POM LOG: Initializing Train Search
Results structural verification sequence
```

```
2026-05-23 11:24:03,955 - INFO - POM LOG: TRAIN SEARCH RESULTS CONTENT
FRAMES SUCCESSFULLY VALIDATED AND STABILIZED
2026-05-23 11:24:03,963 - INFO - Train listing results page successfully
validated.
2026-05-23 11:24:03,963 - INFO - STEP PASSED: the train listing results
page should be loaded - CAPTURING SCREENSHOT
2026-05-23 11:24:04,200 - INFO - Screenshot saved at:
reports/screenshots/PASS_the_train_listing_re_20260523_112403.png
2026-05-23 11:24:04,476 - INFO - POM LOG: Executing short-circuit alias
method call -> apply_ac_filter
2026-05-23 11:24:04,477 - INFO - POM LOG: Dispatching interaction
requests for Air Conditioning (AC) class layout filter
2026-05-23 11:24:06,377 - INFO - POM LOG: AC FILTER CONFIRMED VIA NATIVE
ELEMENT POINTER CLICK ACTION
2026-05-23 11:24:09,385 - INFO - POM LOG: Executing short-circuit alias
method call -> apply_available_filter
2026-05-23 11:24:09,385 - INFO - POM LOG: Dispatching interaction
requests for Seat Availability state verification filters
2026-05-23 11:24:10,541 - INFO - POM LOG: AVAILABILITY FILTER CONFIRMED
VIA NATIVE ELEMENT POINTER CLICK ACTION
2026-05-23 11:24:13,542 - INFO - STEP PASSED: User applies the AC and
Available filters - CAPTURING SCREENSHOT
2026-05-23 11:24:13,714 - INFO - Screenshot saved at:
reports/screenshots/PASS_User_applies_the_AC_20260523_112413.png
2026-05-23 11:24:13,913 - INFO - POM LOG: Parsing layout map structures
to intercept first available active inventory card element
2026-05-23 11:24:16,943 - INFO - POM LOG: ELEMENT STRING TARGET VALUE
VERIFICATION COMPLETED ---> Content payload read: 'Available 177'
2026-05-23 11:24:18,689 - INFO - POM LOG: TICKET INVENTORY INTERACTION
LINK RESOLVED VIA NATIVE CLICK MAPPING
2026-05-23 11:24:24,035 - INFO - STEP PASSED: User selects the first
available train ticket - CAPTURING SCREENSHOT
2026-05-23 11:24:30,812 - INFO - Screenshot saved at:
reports/screenshots/PASS_User_selects_the_fir_20260523_112424.png
2026-05-23 11:24:31,042 - INFO - POM LOG: Successfully verified that the
train details box is loaded.
2026-05-23 11:24:31,043 - INFO - STEP PASSED: the specific train details
page should be loaded - CAPTURING SCREENSHOT
2026-05-23 11:24:31,181 - INFO - Screenshot saved at:
reports/screenshots/PASS_the_specific_train_d_20260523_112431.png
2026-05-23 11:24:31,355 - INFO - POM LOG: Preparing workspace interface
parameters to dispatch interactive link for 'Add Traveller' trigger link
2026-05-23 11:24:33,903 - INFO - POM LOG: NEW MODAL FRAME SCHEDULER
LAYOUT DISPATCHED AND DETECTED IN WEBVIEW VIEWPORT
2026-05-23 11:24:33,905 - INFO - POM LOG: Injecting passenger profile
data matrices -> Data payload maps: [NAME: Harsh | AGE: 25 | GENDER:
Male]
2026-05-23 11:24:36,533 - INFO - POM LOG: Successfully entered passenger
name
```

2026-05-23 11:24:37,237 - INFO - POM LOG: Successfully entered passenger age

2026-05-23 11:24:39,059 - INFO - POM LOG: Gender dropdown expanded

2026-05-23 11:24:39,741 - INFO - POM LOG: Successfully selected gender option: Male

2026-05-23 11:24:42,301 - INFO - POM LOG: MODAL WORKSPACE FORM DATA SUBMITTED AND RETAINED IN COMPONENT CONTEXT SYSTEM

2026-05-23 11:24:42,301 - INFO - STEP PASSED: User opens the add traveller modal and enters passenger details from Excel - CAPTURING SCREENSHOT

2026-05-23 11:24:42,406 - INFO - Screenshot saved at: reports/screenshots/PASS\_User\_opens\_the\_add\_t\_20260523\_112442.png

2026-05-23 11:24:42,523 - INFO - POM LOG: Processing configuration layouts to resolve IRCTC Identity verification values for: abhinavy918

2026-05-23 11:24:53,077 - INFO - POM LOG: Focus verification layer missing modal elements. Injecting forced execution click on trigger selector frame.

2026-05-23 11:25:05,189 - INFO - POM LOG: Resolving focusing events natively onto selected target input frames inside system layers

2026-05-23 11:25:06,134 - INFO - POM LOG: DATA CHARACTERS COMMITTED IN MODAL BUFFER STACKS SUCCESSFULLY

2026-05-23 11:25:10,196 - INFO - POM LOG: IDENTITY MODAL CONTEXT SUBMITTED TO BACKEND CONTROLLER LAYER

2026-05-23 11:25:10,196 - INFO - STEP PASSED: User enters IRCTC username from Excel - CAPTURING SCREENSHOT

2026-05-23 11:25:10,294 - INFO - Screenshot saved at: reports/screenshots/PASS\_User\_enters\_IRCTC\_us\_20260523\_112510.png

2026-05-23 11:25:10,403 - INFO - POM LOG: Loading network input values for identity metadata entries -> [EMAIL: example@test.com | MOBILE: 9876543210]

2026-05-23 11:25:11,345 - INFO - POM LOG: CONTACT SPECIFIC FIELDS COMMITTED TO FRONTEND STORAGE MATRIX SUCCESSFULLY

2026-05-23 11:25:11,346 - INFO - STEP PASSED: User fills contact details from Excel - CAPTURING SCREENSHOT

2026-05-23 11:25:11,446 - INFO - Screenshot saved at: reports/screenshots/PASS\_User\_fills\_contact\_d\_20260523\_112511.png

2026-05-23 11:25:11,569 - INFO - POM LOG: Initializing location verification routines for zero-fee booking cancellation insurance option mapping

2026-05-23 11:25:13,122 - INFO - POM LOG: BOOKING PROTECTION FEE SELECTION PACKETS RESOLVED AND MARKED SUCCESSFULLY

2026-05-23 11:25:13,122 - INFO - POM LOG: Validating legal Terms and Confirmation compliance components

2026-05-23 11:25:13,742 - INFO - POM LOG: COMPLIANCE CHECKBOX VALIDATED AND MARKED TRUE

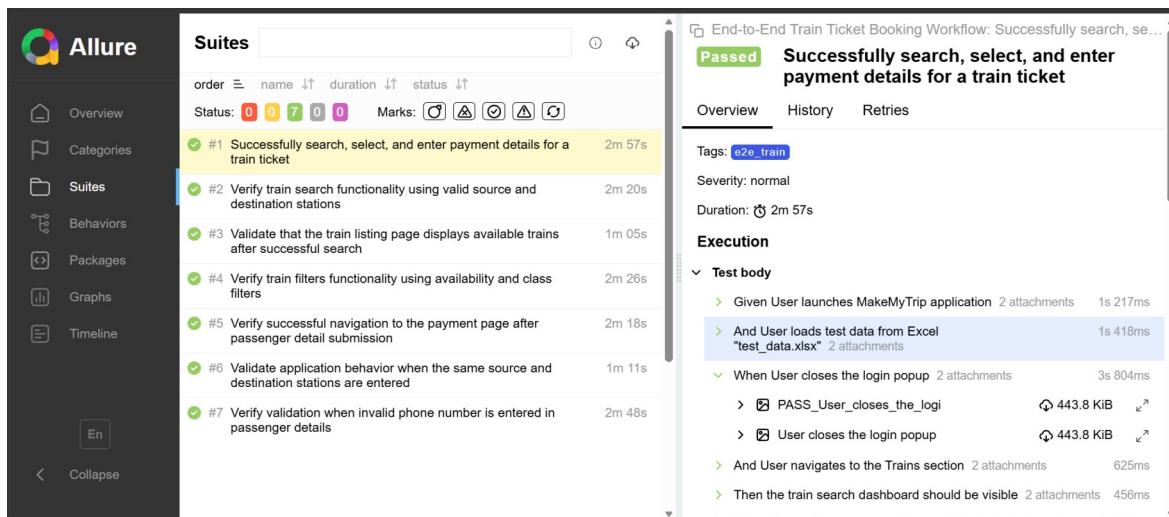
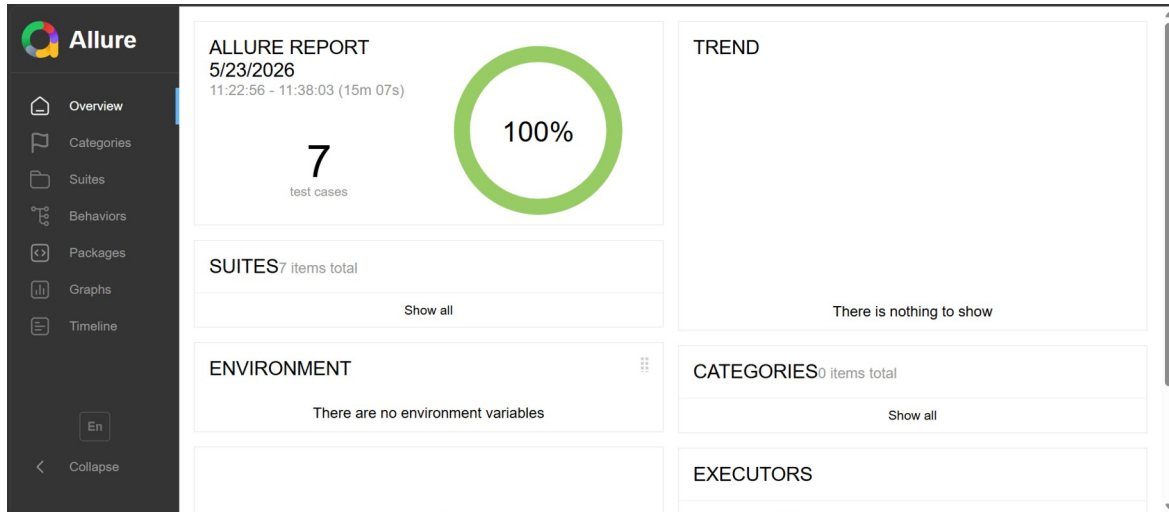
2026-05-23 11:25:14,743 - INFO - STEP PASSED: User selects cancellation insurance and accepts mandatory terms - CAPTURING SCREENSHOT

2026-05-23 11:25:14,856 - INFO - Screenshot saved at: reports/screenshots/PASS User selects cancell 20260523 112514.png



```
2026-05-23 11:25:14,973 - INFO - POM LOG: Routing data validation sets to
locate final form submission checkout link action elements
2026-05-23 11:25:15,533 - INFO - POM LOG: PRIMARY CHECKOUT INTENT
FORWARDED UNTO SETTLEMENT ENGINE GATEWAYS
2026-05-23 11:25:15,534 - INFO - STEP PASSED: User proceeds to payment -
CAPTURING SCREENSHOT
2026-05-23 11:25:15,624 - INFO - Screenshot saved at:
reports/screenshots/PASS_User_proceeds_to_pay_20260523_112515.png
2026-05-23 11:25:15,725 - INFO - POM LOG: Parsing core system layouts to
locate credit/debit transaction card workspace tabs
2026-05-23 11:25:35,482 - INFO - POM LOG: TRANSACTION INTERFACE TAB
SWITCH EVENT DETECTED AND COMPLETED SUCCESSFULLY
2026-05-23 11:25:37,506 - INFO - Payment page validated and credit card
option selected.
2026-05-23 11:25:37,508 - INFO - STEP PASSED: the payment page should be
loaded and credit card option selected - CAPTURING SCREENSHOT
2026-05-23 11:25:37,700 - INFO - Screenshot saved at:
reports/screenshots/PASS_the_payment_page_sho_20260523_112537.png
2026-05-23 11:25:37,949 - INFO - POM LOG: Launching mock value data
injections across protected billing credential inputs
2026-05-23 11:25:38,316 - INFO - POM LOG: ENCRYPTED DATA INPUT INJECTION
COMMITTED -> [FIELD: CARD NUMBER]
2026-05-23 11:25:39,462 - INFO - POM LOG: ENCRYPTED DATA INPUT INJECTION
COMMITTED -> [FIELD: EXPIRY MONTH -> Value: 12]
2026-05-23 11:25:40,117 - INFO - POM LOG: ENCRYPTED DATA INPUT INJECTION
COMMITTED -> [FIELD: EXPIRY YEAR -> Value: 28]
2026-05-23 11:25:40,713 - INFO - POM LOG: ENCRYPTED DATA INPUT INJECTION
COMMITTED -> [FIELD: CVV SECURITY PATCH]
2026-05-23 11:25:41,304 - INFO - POM LOG: ENCRYPTED DATA INPUT INJECTION
COMMITTED -> [FIELD: CARDHOLDER NAME IDENTITY]
2026-05-23 11:25:42,305 - INFO - POM LOG: MOCK TRANSACTION SEQUENCES
EXECUTED AND CONCLUDED WITHOUT THROWING UNHANDLED STACK BOUNDS.
2026-05-23 11:25:45,305 - INFO - STEP PASSED: User enters credit card
details from Excel - CAPTURING SCREENSHOT
2026-05-23 11:25:45,437 - INFO - Screenshot saved at:
reports/screenshots/PASS_User_enters_credit_c_20260523_112545.png
2026-05-23 11:25:45,623 - INFO - POM LOG: Payment Continue button
successfully validated as active. Test safely completed.
2026-05-23 11:25:45,623 - INFO - STEP PASSED: the payment continue button
should become active - CAPTURING SCREENSHOT
2026-05-23 11:25:45,739 - INFO - Screenshot saved at:
reports/screenshots/PASS_the_payment_continue_20260523_112545.png
2026-05-23 11:25:45,921 - INFO - ===== CLOSING SCENARIO:
Successfully search, select, and enter payment details for a train ticket
=====
2026-05-23 11:25:53,144 - INFO - BROWSER CLOSED SUCCESSFULLY
```

### 4B.3 Allure Report — BDD



## 5. TEST REPORT

---

### 5A. Selenium + pytest — Test Report

#### *Execution Summary*

| Metric             | Value                     |
|--------------------|---------------------------|
| Total Tests        | 7                         |
| Tests Passed       | 7                         |
| Tests Failed       | 0                         |
| Tests Skipped      | 0                         |
| Pass Rate          | 7 / 7 = 100 %             |
| Execution Duration | ~4 minutes                |
| Report Format      | Allure HTML + pytest-html |

#### *Accuracy Metric*

Pass Rate = Total Passed / Total Tests × 100 = 7 / 7 × 100 = 100.00 %

---

### 5B. BDD behave — Test Report

#### *Execution Summary*

| Metric             | Value                       |
|--------------------|-----------------------------|
| Total Scenarios    | 7                           |
| Scenarios Passed   | 7                           |
| Scenarios Failed   | 0                           |
| Scenarios Skipped  | 0                           |
| Pass Rate          | 7 / 7 = 100 %               |
| Feature Files      | 2                           |
| Execution Duration | ~4.5 minutes                |
| Report Format      | Allure HTML (allure-behave) |

### *Accuracy Metric*

Pass Rate = Total Passed / Total Scenarios  $\times 100 = 7 / 7 \times 100 = 100.00 \%$